

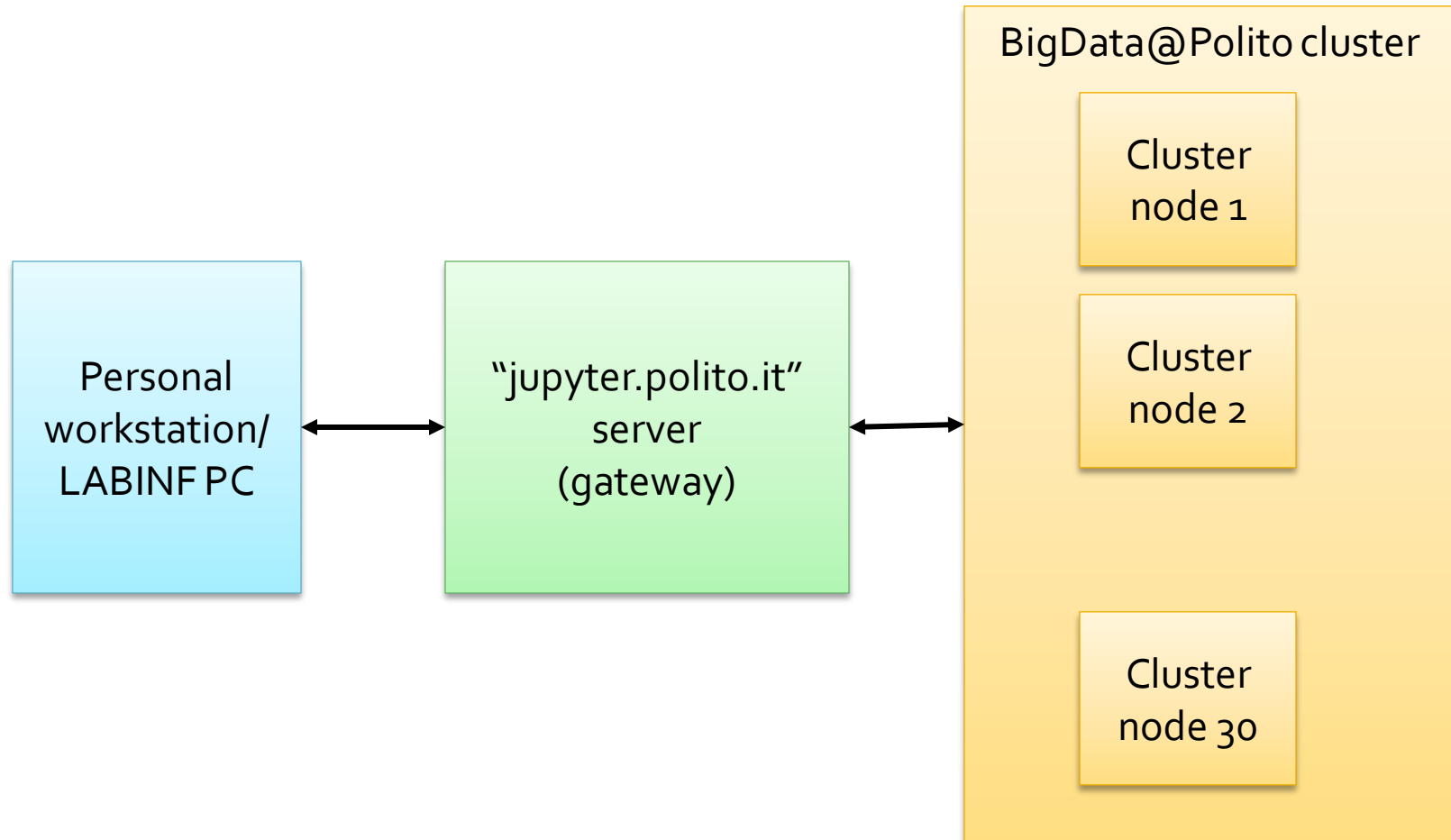
Execution of MapReduce applications

The BigData@Polito environment + Jupyter

The BigData@Polito environment

- The BigData@Polito cluster has
 - A set of servers running Hadoop
 - An **Access Gateway** server used to interact with the Hadoop cluster
 - Submit jobs/execute MapReduce applications
 - Submit hdfs commands
 - Analyze the log files from command line

The BigData@Polito environment



The BigData@Polito environment – Execute an application (1)

- Execute a MapReduce Application on the cluster (i.e., submit a MapReduce job on the cluster)
 - Log into jupyter
 - <https://jupyter.polito.it>
 - Copy the jar file containing your application from your personal workstation (or the workstation of LABINF) in the “local” file system of the gateway
 - Drag and Drop from your PC to the jupyter web page
 - Copy the input data of your application from the “local” file system of your personal workstation in the HDFS file system of the cluster
 - Drag and Drop from your PC to the jupyter web page and then use the HDFS command
 - Or use the HUE web interface
 - <https://hue.polito.it>

The BigData@Polito environment – Execute an application (2)

- Open a terminal in Jupyter
- Use the hadoop command from the opened terminal to submit the job
 - Specify the name of the jar file, the name of the input (HDFS) data, the name of output folder, the parameters/arguments of the application
 - Example

```
hadoop jar Exercise1-1.0.0.jar  
it.polito.bigdata.hadoop.exercise1.DriverBigData 2  
ex1_data ex1_out
```

The BigData@Polito environment – Execute an application (2)

- Open a terminal in Jupyter

This command execute an application on the cluster

- Exercise1-1.0.0.jar: jar file containing the code of the MapReduce application
- it.polito.bigdata.hadoop.exercise1.DriverBigData: driver class
- This application has three parameters
 - Number of instances of the reducer
 - Input folder
 - Output folder

- Example



```
hadoop jar Exercise1-1.0.0.jar  
it.polito.bigdata.hadoop.exercise1.DriverBigData 2  
ex1_data ex1_out
```